TRANSMISSION LINES

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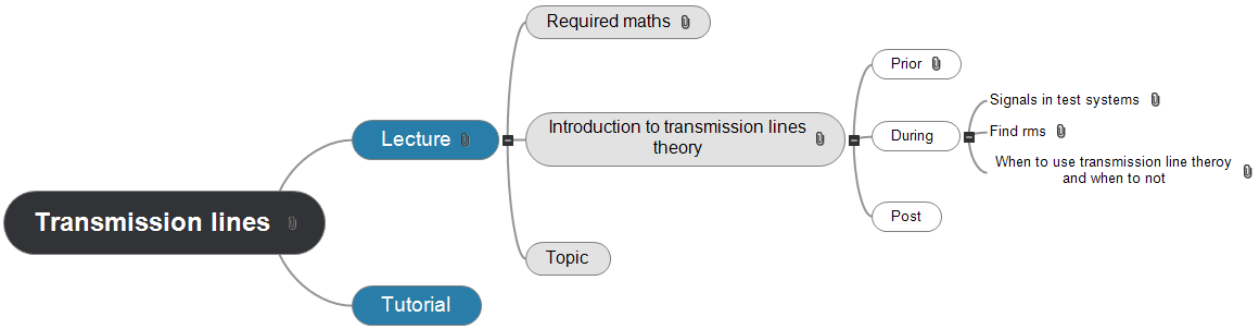

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1 LECTURE

All of abduls equations weill be included as pictures or if im confused as to how to wrtite it then use the photo

I cant read joined up well so any help understanding is great

1.1 REQUIRED MATHS

Complex numbers

1.2 INTRODUCTION TO TRANSMISSION LINES THEORY

As an intro lecture this coverd required maths and getting started

1.2.1 PRIOR

Will now be post and this will be the same for most as i have no time wooooooo

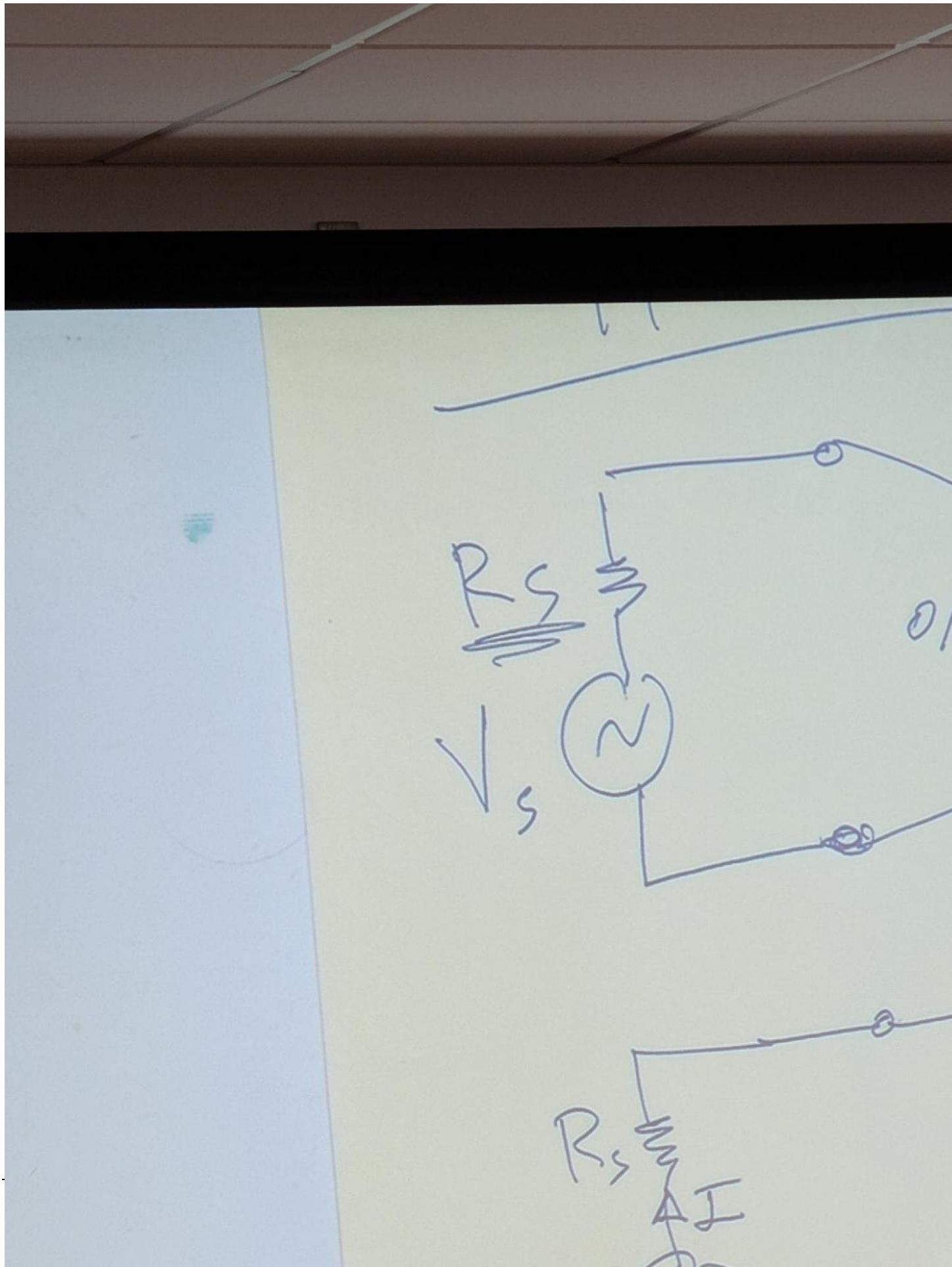
1.2.2 DURING

1.2.2.1 SIGNALS IN TEST SYSTEMS

To find I - vs / rs + zl

Vl = vs / rs + zl * zl

Vl = vs * zl/2s + zl


R_S
V_S
R_S
ΔI

Vz = vs * r2/r1+r2

Vi = vs * r1/r1+r2

$$V_L = V_s \times \frac{Z_L}{Z_s}$$

Voltage Dividen
Method

We assume that transmission lines have no resistance to get the numbers so far so from here on we need to find the transmission lines(these are the super perfect conductiors that can never excist)

To find the power of p2

P = V * I

power = vl *I

Power = v^2/z

Power I^2/z

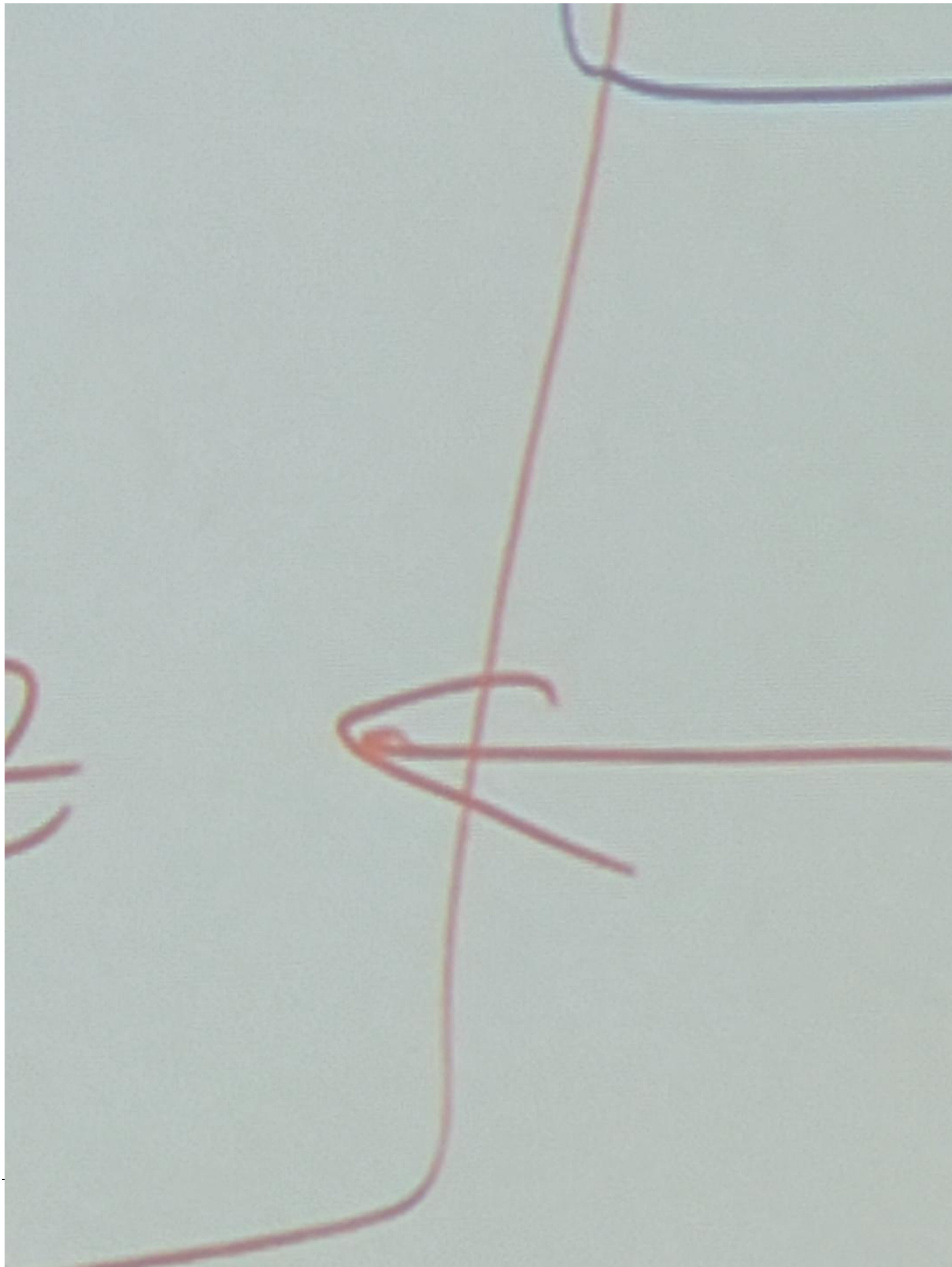
$$V_L = V_s \times \frac{}{R}$$

Voltage Divide
Metho

Power = V × I

Power load = V_L ×

Vp is peak voltage vp . P is peak to peak voltage



On osilliscopes you can read peak and peak to peak volatage on avanced ones you can read more

1.2.2.2 FIND RMS

V rms = vp/sqrt(2) FOR SINE ONLY others use different rms values

Rms actualy =

$$V_{rms} = \frac{V_P}{\sqrt{2}}$$

$$\frac{1}{T}\int^{T} f($$

$$V_{rms} = \frac{V_P}{\sqrt{2}}$$

$$\frac{1}{T}\int_0^T f(t)^2 \, d$$

To find power =

Power = vrms * Irms

Cos(theta)

$$\frac{1}{T}\int_0^T f(t)^2$$

$$\text{Power} = V_{rms} \times$$

1.2.2.3 WHEN TO USE TRANSMISSION LINE THEROY AND WHEN TO NOT

Upto this point we use krichovs laws and similar constants but not transmission lines (transmission lines still use these theorys just have more detail)

Electrical length measn the number of wave lengths for a particular physical length

Take lp

Lpe = Lp / wavelength

$$\frac{1}{T}\int_0^T f(t)$$

Power = rms

Wavelength is distance traveled in a period its unit is m mm cm so on

D = v* T

v= is propagation velocity

C = for free space in air radio wave travel at speed of light 3*10^8 m/s

T = Time is 1 second

Er = relative permeability of the material

So vp = c/er

Wavelength = vp/frequency so 1hz = 1 second so if it happens 1 time a second its a 1 hz freq

1.2.3 POST

1.3 TOPIC

2 TUTORIAL